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DARSHAN UNIVERSITY

Department of Computer Science & Engineering

PROJECT REPORT

Gujarati Dastavej AI Valuator

A Vision-Language AI System for Extracting, Verifying, and Valuing Gujarati Property Documents

Submitted for consideration towards the

Dewang Mehta IT Awards

Submitted by

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Enrollment No: 23010101083

Guide / Head of Department

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B.Tech — Computer Science & Engineering | Academic Year 2025–26

Certificate of Project Completion

This is to certify that the project report entitled **“Gujarati Dastavej AI Valuator”** has been carried out by **Devansh Ganatra (Enrollment No: 23010101083)**, a student of B.Tech — Computer Science & Engineering, Darshan University, in partial fulfillment of the requirements for submission to the Dewang Mehta IT Awards, during the academic year 2025–26.

The work presented in this report is an authentic record of the candidate's own work, carried out under the guidance stated below, and to the best of our knowledge has not been submitted elsewhere for any other award.

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Prof. Jayesh Vagadiya
Guide / Head of Department
Dept. of Computer Science & Engineering

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[External Examiner]
Dewang Mehta IT Awards Panel
[Placeholder — to be signed]

Declaration

I, Devansh Ganatra, Enrollment No: 23010101083, hereby declare that the project report entitled "Gujarati Dastavej AI Valuator", submitted for the Dewang Mehta IT Awards, is an original record of work carried out by me under the guidance of Prof. Jayesh Vagadiya, Department of Computer Science & Engineering, Darshan University.

I further declare that the system architecture, source code, testing methodology, and results presented in this report reflect the actual state of the working application at the time of submission, and that no functionality has been misrepresented as complete where it is still in progress (see Chapter 8, Future Scope, for work planned but not yet implemented).

This work has not been submitted, in part or in full, to any other university, board, or competition for the award of any degree or recognition.

Devansh Ganatra
Enrollment No: 23010101083

Acknowledgement

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I would also like to acknowledge the organizers of the Dewang Mehta IT Awards for creating a platform that encourages students to build technology solutions with genuine, real-world relevance rather than purely academic exercises.

Finally, I thank my family and peers for their continuous support and patience during the many iterations of testing and refinement this project went through.

Abstract

Property valuation in Gujarat begins with a document almost no automated tool can read: the Dastavej, a legal property record written in Gujarati script, often scanned at inconsistent quality, and structured in ways that vary from office to office. Every bank officer, valuer, and legal professional who has ever processed one has done the same thing — read it manually, and typed what they found into a form by hand. This project automates that first, most repetitive step, without ever pretending the AI's reading is final.

The Gujarati Dastavej AI Valuator uses a vision-language model (Google Gemini) to read property documents the way a person would — by looking at the page, not by running classical OCR against a dictionary of Latin characters. It extracts structured data across four real Gujarat land-record formats (Dastavej, 7/12 Extract, Property Card, and Index-II), flags any field the model itself was unsure about, and standardizes every extracted value into a professional English report — the language Gujarat's banks and valuation offices already expect regardless of the source document's script — while giving the user a direct link to manually verify the record against the Gujarat government's own AnyROR portal.

The system also enforces a real statutory boundary: Certified Land and Building Valuers in India are not authorized to value agricultural land, so the valuation calculator is automatically disabled — not merely hidden — for the two document types tied to agricultural or registration-only records. This is treated as an independent regulatory fact rather than a claim made by any one reviewer; the project's field validation review (Appendix B) is itself explicitly scoped by its author to matters within a Civil Engineer's professional domain, and does not extend to agricultural-land valuation authority, which sits with a different technical authority.

Unlike generic document-AI platforms that charge a monthly minimum before processing a single page and have no notion of Gujarati script or Indian land-record formats, this system runs at a marginal cost of a fraction of a rupee per document. More importantly, it was built and hardened the way production software should be: real bugs were found through deliberate adversarial testing — a crash on two of four document types, and a silent miscalculation on the Hectare-Are-Sq.Mtr area format used on every 7/12 extract — and fixed before being called complete, not discovered by a user in the field.

This report documents the system's architecture, the engineering decisions behind it, the testing discipline used to harden it, and its relevance to financial inclusion for Gujarat's rural landowners and the bank officers and valuers who serve them.

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Chapter 1: Introduction

1.1 Background & Motivation

A Dastavej is a registered Gujarati property document — a sale deed, gift deed, or partition deed — and it is the starting point of nearly every property transaction, bank loan, and legal verification in Gujarat. These documents are written in Gujarati script, frequently scanned from decades-old paper records, and structured differently across Sub-Registrar offices and time periods.

Every property valuer, bank loan officer, and legal professional handling these documents today does the same manual task: read the Gujarati text, identify the owner name, survey number, area, and boundaries, and type that information into a valuation form. This is slow, repetitive, and error-prone precisely where errors matter most — a misread survey number or area figure can misdirect an entire loan or legal filing.

Generic document-AI platforms exist, but almost none of them are built with Gujarati script or Indian land-record formats in mind (see Chapter 2). This leaves a specific, real gap: a tool that reads Gujarati property documents the way a person does, and hands back structured, verifiable data — not a black box, and not a replacement for a human valuer's judgment.

1.2 Problem Statement

To design and implement an AI-assisted system that extracts structured, verifiable data from Gujarati property documents (Dastavej, 7/12 Extract, Property Card, and Index-II), surfaces its own uncertainty rather than hiding it, computes a first-pass property valuation from the extracted data, and produces bank- and legal-ready reports — without requiring the human reviewer to trust the AI blindly.

1.3 Objectives

- Automate extraction of structured data from Gujarati property documents using a vision-language model, eliminating manual re-typing.
- Support the four Gujarat land-record document types a valuation actually depends on: Dastavej, 7/12 Extract, Property Card, and Index-II.
- Surface the model's own confidence per field, so a human reviewer knows exactly which values to double-check rather than trusting the output uniformly.
- Compute a first-pass estimated valuation directly from extracted area and a user-supplied rate, handling real-world area formats (including the Hectare-Are-Sq.Mtr notation used on 7/12 extracts).
- Generate export-ready reports — a professional, English-standardized PDF and a structured Excel workbook — suitable for banks and legal professionals.
- Enforce the real statutory boundary on which document types a Certified Land and Building Valuer may actually value, rather than computing an estimate regardless of legal authority to do so.
- Provide a legitimate, non-intrusive path to cross-verify extracted data against the Gujarat government's own AnyROR land-record portal.

1.4 Scope of the Project

The current system covers single-document processing across four document types, with a human-in-the-loop review step before any export. It is deliberately scoped to avoid two things that would be inappropriate for a student project handling legal/financial documents: it does not compute a legally binding valuation (it produces a first-pass estimate for the reviewer to adjust), and it does not scrape or automate the Gujarat government's land-record portals — verification against AnyROR is a manual, one-click deep link, not an automated lookup (see Chapter 4.5 for why this boundary was chosen deliberately).

Batch processing, a persistent history database, multi-user accounts, and camera-capture input are planned but not yet implemented; these are documented honestly in Chapter 8 as future scope rather than presented as delivered features.

1.5 Report Structure

Chapter 2 examines the existing manual process and comparable commercial tools, and where they fall short for this specific problem. Chapter 3 presents the proposed system architecture. Chapter 4 details the implementation, including the specific engineering decisions and edge cases handled. Chapter 5 documents the testing methodology and the real bugs it caught. Chapter 6 discusses the project's social and financial-inclusion relevance. Chapter 7 presents results. Chapter 8 concludes and outlines future work.

Chapter 2: Existing System & Market Analysis

2.1 The Manual Process Today

Without a tool like this, processing one Dastavej for valuation purposes typically involves: a human reading the scanned Gujarati document on-screen, manually transcribing owner name, survey number, area, and boundary details into a spreadsheet or form, converting units (square meters to square feet) by hand, and separately looking up any government reference values needed for context. For a bank processing dozens of agricultural loan applications a week, this is a meaningful, recurring time cost, and every manual transcription step is a place a number can be mistyped.

2.2 Generic Document-AI Platforms

Several commercial intelligent document processing (IDP) platforms exist — Google Document AI, Docsumo, and Nanonets among the most established. All three are capable, general-purpose tools, but none of them are built for this problem specifically:

- None offer a pre-built extractor for Gujarati script or Indian state-specific land-record formats (7/12, Property Card, Index-II) — each would require custom model training from scratch on a format none of them have seen before.
- All three charge a recurring monthly minimum or per-page fee that applies whether or not the documents involved are ones they can actually read well, before a single successful extraction happens.
- None compute a property valuation — they are generic extraction engines, not domain tools for this use case.

2.3 Cost Comparison

The chart below shows an approximate, indicative cost per document across the compared platforms. Figures are drawn from each platform's current public pricing as of mid-2026 and are necessarily approximate — actual cost depends on document length, plan tier, and negotiated enterprise rates.

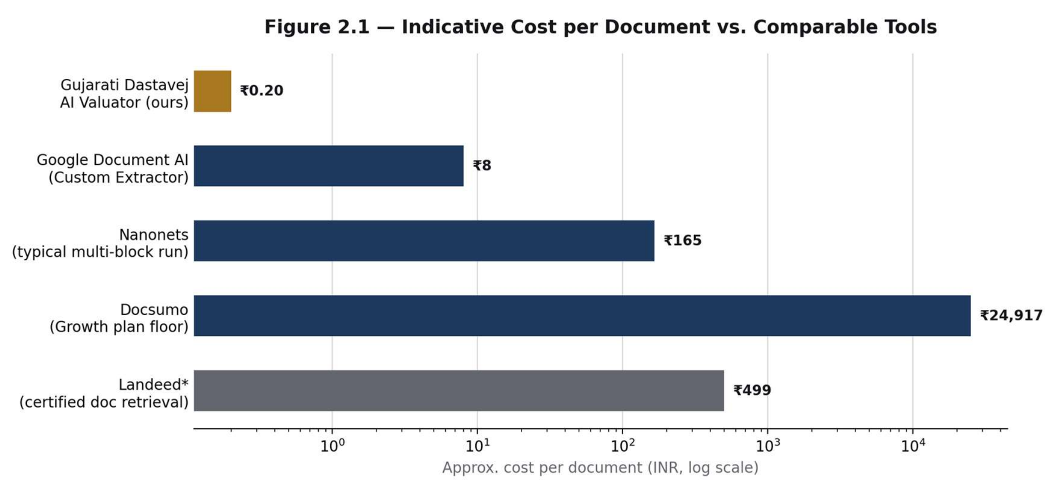


Figure 2.1 — Indicative cost per document. Log scale used because the range spans roughly three orders of magnitude.

Platform	Typical Cost Basis	Handles Gujarati Land Records?
Gujarati Dastavej AI Valuator (ours)	~₹0.20/document (Gemini 2.5 Flash token cost at this scale)	Yes — purpose-built
Google Document AI (Custom Extractor)	~$30 per 1,000 pages, plus custom model training/integration effort	No pre-built extractor
Nanonets	~$0.30 per AI processing block; a typical multi-block document run is often cited around $1–$2 end-to-end	No pre-built extractor
Docsumo	Entry (Growth) plan starts at $299/month for 1,000 pages; per-page rate commonly cited at $0.30–$0.50	No pre-built extractor
Landeed*	₹199–₹699 per certified document (retrieval fee, not extraction)	N/A — different job entirely

**Landeed retrieves the official government document for a fee — it does not extract structured data or compute a valuation from it, so it is not a like-for-like competitor, but is included since it is the closest existing tool in the Gujarat land-record space.*

2.4 The Gap This Project Fills

No existing tool combines (a) native support for Gujarati-script land-record documents, (b) a computed first-pass valuation from extracted data, and (c) a near-zero marginal cost per document that makes it viable for individual valuers and small offices rather than only large enterprise budgets. This is the specific, validated gap the Gujarati Dastavej AI Valuator addresses.

Chapter 3: Proposed System

3.1 System Overview

The system treats each property document as an image, not a text file — because that is what a scanned Gujarati Dastavej actually is. A page image is sent, together with a type-specific instruction prompt, to a vision-language model, which returns structured data as JSON along with a confidence rating per field. That output is never final: it lands in an editable review form where the human reviewer sees exactly which fields the model flagged as uncertain, corrects anything necessary, and only then triggers the valuation calculation and export.

3.2 Why a Vision-Language Model, Not Classical OCR

Classical OCR (optical character recognition) works by matching pixel shapes against a trained dictionary of characters, then applying rule-based or statistical language correction. This works reasonably well for clean, printed Latin-script text. It performs poorly on Gujarati property documents for three compounding reasons: the script itself has complex conjunct characters that classical OCR engines are rarely trained on well, the documents are frequently low-quality scans of old paper records, and the layout is semi-structured — field labels and values are positioned inconsistently across different offices and years.

A vision-language model (VLM) like Google Gemini processes the page holistically — combining a vision encoder (interpreting the image) with a language model (interpreting context and meaning) in a single pass. This lets it use surrounding context to disambiguate an unclear character the way a human reader would ("this is probably the survey number because of where it sits on the page and what precedes it"), rather than matching isolated character shapes in a vacuum. This is the core technical justification for the architecture: the problem is not "read the characters," it is "understand the document," and that requires a model that reasons over layout and language together.

3.3 System Architecture

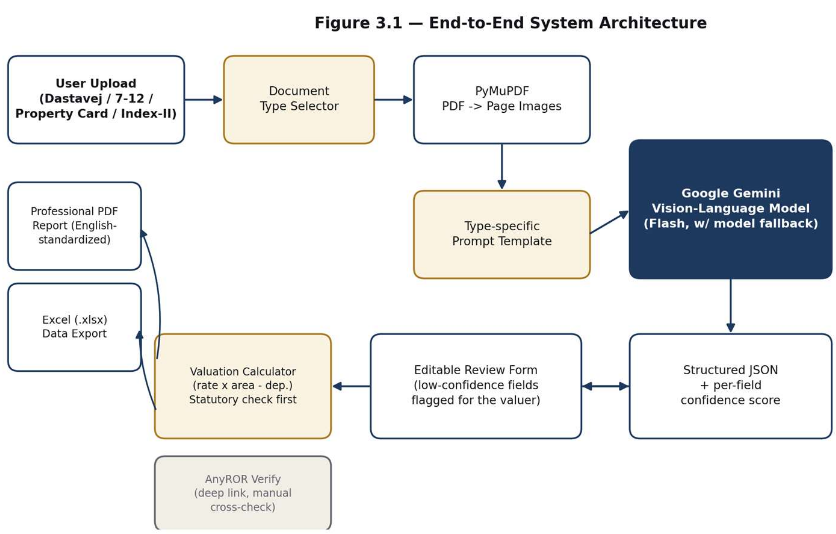


Figure 3.1 — End-to-end system architecture. Every AI output passes through a human review step before any export is generated.

The pipeline has five stages: (1) the user selects a document type and uploads a PDF or scanned page, (2) PyMuPDF converts each page into a high-resolution image, (3) a type-specific prompt and the page images are sent to Gemini, which returns structured JSON with a confidence score for every field, (4) the extracted data populates an editable form where low-confidence fields are visibly flagged, and only after human review does (5) the valuation calculator run — if the document type is statutorily eligible for one — and the final English-standardized PDF and Excel exports get generated.

3.4 Multi-Document-Type Design

A single valuation case in Gujarat rarely involves only a Dastavej — a 7/12 Extract, Property Card, or Index-II is often needed alongside it to cross-check ownership and area. The system supports all four as first-class document types, each with its own extraction schema and prompt, while sharing one underlying pipeline (model fallback logic, defensive JSON parsing, confidence scoring, and image conversion) so that adding support for a document type does not mean rebuilding the system underneath it.

Figure 3.2 — One Pipeline, Four Gujarat Land-Record Document Types

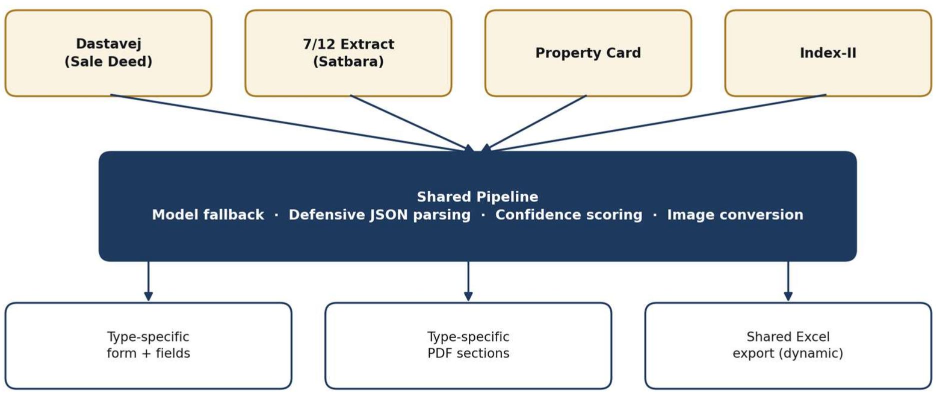


Figure 3.2 — One shared pipeline serves four independently-schemed document types.

3.5 Technology Stack

Layer	Technology	Purpose
Web interface	Streamlit (Python)	Rapid, form-driven UI for upload, review, and export
PDF processing	PyMuPDF (fitz)	Converts PDF pages into high-resolution images for vision input
Extraction engine	Google Gemini (Flash family, vision-language model)	Reads document images and returns structured JSON with confidence scores
Data handling	Pandas	Structures extracted data for the Excel export
PDF report generation	fpdf2 (core Helvetica font)	Produces a professional, English-standardized PDF valuation report

Layer	Technology	Purpose
Excel export	openpyxl (via Pandas)	Structured, unicode-native spreadsheet export
Image handling	Pillow (PIL)	In-memory image processing between PyMuPDF and Gemini

3.6 Module Description

Extraction Engine

Handles PDF-to-image conversion, model selection with automatic fallback across available Gemini Flash models, prompt dispatch per document type, and defensive parsing of the model's JSON response (including recovery from markdown-fenced or prose-wrapped responses).

Confidence-Flagged Review Form

Every extracted field carries a model-reported confidence rating. Fields marked low or medium confidence are visibly flagged in the review form with a short caption, directing the reviewer's attention exactly where a misread is most likely — rather than treating every field as equally trustworthy.

Valuation Calculator

Computes an estimated property value from the extracted area and a reviewer-supplied rate per square foot, with a simple age-based depreciation adjustment. It runs live as the reviewer adjusts inputs, using Streamlit's fragment-scoped reruns so this recalculation does not force the rest of the page (including the document preview) to re-render on every keystroke.

Professional English Export

Every extracted value is standardized into professional English before it reaches the export stage, regardless of the source document's script (see Chapter 4.3 for the reasoning behind this and the edge cases it introduced).

Statutory Eligibility Check

Before the valuation calculator renders at all, the system checks whether the selected document type is one a Certified Land and Building Valuer is actually authorized to value. If not, the calculator is replaced with an explanatory notice and the exported value is hardcoded accordingly (see Chapter 4.6).

AnyROR Verification Link

For the two document types tied to a specific survey number or city survey number (7/12 Extract and Property Card), the form includes a one-click link to the relevant section of the Gujarat government's AnyROR portal, so a reviewer can manually cross-check the AI's extraction against the live government record. This is a static deep link only — see Chapter 4.5 for why automated scraping was deliberately ruled out.

3.7 Interface & Reliability Polish

Beyond the extraction pipeline itself, several interface-level issues were resolved to make the tool usable in a real workflow rather than only in a demo: a resizable, skeleton-style loading state replaces the generic spinner while a document is processing; the file upload widget is force-rebuilt on reset so a cleared session cannot leave stale file state behind; and the resolution used to convert PDF pages into images now scales down automatically for longer documents, preventing out-of-memory failures on large scanned files rather than applying one fixed setting regardless of document length.

Chapter 4: Implementation Highlights

This chapter documents the specific engineering decisions and edge cases that shaped the working system — not a line-by-line code walkthrough, but the design choices that mattered and why.

4.1 Extraction Schema Across Four Document Types

Each document type has its own field schema, matched to what that specific Gujarat land-record format actually contains. A summary of the highest-value fields per type:

Document Type	Key Fields Extracted	Primary Use
Dastavej (Sale Deed)	Owner name, father/husband name, survey number, area, four boundary directions, document number	Ownership transfer verification
7/12 Extract (Satbara)	Village/Taluka/District, survey & block number, khata number, owner(s) & share, tenure type, land type, mutation entries, encumbrance/loan notation	Agricultural land ownership & tenure
Property Card	City survey number, ward, sheet/plot number, owner(s) & share, land use type, mutation & encumbrance details, tax assessment number	Urban property ownership
Index-II	Document number, registration date, executant/claimant names, agreement value, Jantri value, stamp duty & registration fee paid	Registration & transaction-value verification

These field lists were deliberately researched against the actual structure of AnyROR- and Garvi-issued documents rather than guessed — an incorrect field name is worse than a missing one, since it would silently fail to extract data a reviewer is relying on.

Two of these fields were refined further after direct feedback from a practicing valuer during field testing, not from synthetic testing (see Chapter 5.2): the Dastavej owner field was initially ambiguous about which party is the current owner, and was corrected to explicitly target the purchaser/buyer rather than the seller; and the boundary fields were being missed on longer documents where the boundary schedule (चतुर्दिशा / चतुर्सीमा) appears on a later page rather than near the main text.

4.2 Per-Field Confidence Flagging

Beyond the field values themselves, the extraction prompt asks the model to return a confidence rating (high / medium / low) for every field, based on how legible the source value was. The review form surfaces this directly: any field rated medium or low confidence gets a visible caption prompting the reviewer to double-check it, rather than presenting every extracted value with equal, unearned authority.

This matters specifically for this document class: numeric fields such as survey number and area are exactly where a silent vision-model misread is most costly, and exactly where a human reviewer's attention needs to be directed first.

4.3 Language Strategy: Gujarati In, Professional English Out

The vision input to the model is always the original Gujarati-script document — that part of the architecture is unchanged from Chapter 3.2. What every extracted field goes through before it reaches the review form or any export, however, is standardization into professional English, regardless of the language on the source page. This is a deliberate design decision made after direct feedback from a practicing valuer during field testing, not a rendering shortcut: valuation reports in Gujarat are routinely expected in English by banks, out-of-station loan processing teams, and legal offices, even when the underlying registered document is in Gujarati. A report that mixed scripts inconsistently would have reintroduced exactly the kind of friction this project set out to remove.

This decision surfaced two real, narrow issues during the transition, both closed before this report:

```
# Issue 1 -- Invisible English text in the PDF
# The Gujarati-script font file was a script-specific subset
# with no Latin glyphs. Once values became English text,
# that font rendered nothing at all for them.
# Fix: render report text with a standard core font, now
# that the output is English throughout.

# Issue 2 -- Rupee symbol crashing the export
value = value.replace(RUPEE_SIGN, 'Rs. ')  # before latin-1 encoding
```

One nuance worth flagging honestly rather than glossing over: personal and place names do not have a meaningful English “translation.” In practice, the model's behavior on these fields is closer to Romanized transliteration — a village name comes out in its standard English spelling, not reinterpreted or invented. This has held up correctly in testing, but proper nouns are exactly the kind of field worth spot-checking by a reviewer rather than trusting by default — which is also precisely why every field, including these, carries its own confidence rating (Section 4.2) rather than being treated as uniformly reliable.

4.4 Valuation Calculation & the Hectare-Are-Sq.Mtr Edge Case

The valuation calculator computes: estimated value = extracted area (converted to sq. ft.) × reviewer-supplied rate per sq. ft., reduced by a straight-line depreciation of 1% per year of property age, capped at 30% total depreciation.

A real edge case surfaced during testing: a 7/12 Extract's Total Area field is conventionally written in Hectare-Are-Sq.Mtr notation — for example “1-23-45”, meaning 1 hectare, 23 are, 45 square meters — not a plain number. A naive parser reading the first number it finds interprets this as simply “1”, silently producing an estimated value near zero with no error shown, on what is a common, expected real-world input format for this document type specifically.

```
# parse_hectare_are_sqm("1-23-45")
# 1 hectare (10,000 sqm) + 23 are (100 sqm each) + 45 sqm
# = 10,000 + 2,300 + 45 = 12,345 sqm  -->  converted to sq. ft.
```

A dedicated parser detects this three-part hyphenated pattern specifically for 7/12 documents and converts it correctly before the standard calculation runs, falling back to plain-number parsing for the other three document types (and for any 7/12 value that does not match the pattern).

4.5 AnyROR Verification — A Deliberately Narrow Feature

The Gujarat government's AnyROR portal has no public API. It is a citizen self-service system — district/taluka/village dropdowns, a survey number, and a CAPTCHA per query — not a developer platform. Building an automated scraper against a CAPTCHA-protected government portal was considered and deliberately rejected: it is fragile against portal changes with no notice or versioning, it cannot run unattended past the CAPTCHA regardless, and automating access to a government computer system in ways not intended by its operator raises real legal exposure under India's IT Act that a student project has no business taking on.

Instead, the system adds a single “Verify on AnyROR” link for the two document types tied to a specific survey/city-survey number. It constructs a URL to the correct section of the official portal and opens it in a new tab — no form submission, no CAPTCHA handling, no scraping, no response parsing. The reviewer still does the actual verification, manually, against the government's own live record. This is arguably more appropriate for the stated audience (bank officers who need defensible, human-verified cross-checks) than a background scrape would have been, and it is a feature that survives a portal redesign rather than breaking on one.

4.6 Statutory Compliance: Why the Calculator Doesn't Always Run

A structured field and a computed valuation are not the same kind of output, and treating them identically would have been a real professional error. Under standard Indian valuation practice, Certified Land and Building Valuers are not statutorily authorised to value agricultural land: a 7/12 Extract almost always represents agricultural land, and an Index-II is a registration summary rather than a property being valued in the same sense as a Dastavej or Property Card. This restriction was implemented as a hard rule in the system rather than left to the reviewer's judgement. It is worth being precise about its source: this is a general rule of the valuation profession, not a claim made by the project's field validation reviewer, whose letter (Appendix B) explicitly limits his comments to document types and valuation methods within a Civil Engineer's domain, and states plainly that matters falling under a different technical or regulatory authority are outside the scope of his review.

The system enforces this directly rather than leaving it to the reviewer's judgment. For these two document types, the Valuation Calculation section does not render at all; a visible notice explains the statutory reason why; and the exported “Estimated Value” field is hardcoded to “N/A (Statutory Restriction)” rather than left blank or silently computed anyway.

```
if document_type in ["7/12 Extract (Satbara)", "Index-II"]:
    st.warning(
        f"Note: {document_type} documents represent agricultural "
        "land or registration summaries. Certified Land and "
        "Building Valuers do not have statutory rights to value "
        "agricultural land."
    )
    return  # calculator section is skipped entirely
```

This is one of the clearer examples in the project of domain input shaping what the system actually does, not just how it looks — and it's also a useful illustration of respecting the limits of that input. The statutory restriction on valuing agricultural land is a general rule of the valuation profession, not a determination the project's field validation reviewer — a Civil Engineer by training — was asked to make about agricultural land itself; a specific opinion on agricultural land properly sits with an Agricultural Engineer, outside a Civil Engineer's domain. His letter (Appendix B) says so directly, which is precisely why it was used as supporting evidence for the parts of the system it does cover — extraction accuracy and general valuation methodology — and not for this specific statutory boundary.

Chapter 5: Testing & Quality Engineering

A document-extraction tool feeding into bank and legal decisions cannot be tested only with clean, cooperative sample data. This project's testing strategy was built specifically to surface the failure modes real users would eventually hit — and it succeeded: every bug listed below was caught by a deliberately adversarial synthetic test document before it ever reached a real user.

5.1 Adversarial Synthetic Test Documents

Rather than relying only on clean, hand-picked example documents, a dedicated script generates synthetic test PDFs designed specifically to break assumptions: area values in every real-world format (plain numbers, Hectare-Are-Sq.Mtr notation, Gujarati numerals), fields left deliberately blank, and one PDF report generated per document type so every export path has a standing, repeatable test rather than relying on manual spot-checks.

5.2 What Testing and Domain Review Caught

Some issues are the kind a well-built synthetic test document will always find eventually. Others are the kind no synthetic test could ever find, because they require actually knowing how a valuer reads a document — not just whether the code runs without error. Both categories surfaced real, specific findings, all closed prior to this report.

Caught by Synthetic Adversarial Testing

Edge Case Identified	How It Was Resolved
Export filenames for 7/12 Extract and Property Card	These two document types use a different identifier field than Dastavej and Index-II. The download logic now selects the correct identifier field per document type automatically.
Area parsing for 7/12's Hectare-Are-Sq.Mtr notation	A dedicated parser was added to correctly convert this three-part notation (e.g. “1-23-45”) into square meters ahead of the valuation calculation, alongside the existing plain-number path already used by the other three document types.

Caught by Domain-Expert Review

A practicing valuer testing the system against real documents surfaced two issues no amount of synthetic testing would have found, since both required domain knowledge a test script does not have:

Edge Case Identified	How It Was Resolved
Owner field capturing the seller instead of the buyer	The prompt originally asked generically for an “owner name,” which a Dastavej itself does not label as such. Corrected to explicitly target the purchaser/buyer — the party who is the actual current owner — matching how a valuer reads the document.
Boundary fields missed on longer, multi-page documents	On documents beyond a few pages, the boundary schedule frequently appears on a later page rather than near the main text. The prompt now explicitly instructs the model to locate that section specifically, rather than assuming it appears early.

The distinction matters: the left-hand column catches whether the software works; the right-hand column catches whether the software is right. Both are necessary, and neither substitutes for the other — which is the same reasoning behind seeking a practicing valuer's review in the first place rather than treating internal testing as sufficient on its own.

This review was subsequently formalised in writing: Harsh Kathrani (Technical Manager, Dot Value Consultancy) provided a signed field-validation letter confirming the specific issues found, the valuation methodology discussed, and his professional opinion on the tool, reproduced in full as Appendix B.

5.3 Efficiency Review

A separate efficiency pass, done before adding batch-processing (which would multiply any per-document inefficiency), identified and addressed three concrete issues rather than general “optimization”:

- The same uploaded PDF was being opened and parsed three separate times (once for a page-count check, once for the preview, once inside the extraction call) — consolidated to a single open per upload.
- The list of available Gemini models was being re-fetched over the network on every single extraction call, even though it rarely changes within a session — now cached for the session.
- The live valuation calculator's rate/age inputs were triggering a full-page rerun (including the document preview panel) on every keystroke — scoped to a Streamlit fragment so only the valuation section re-renders.

5.4 Manual Smoke Test

A short, standing manual checklist (upload, extraction, form population, both export formats, and session reset) is run after every code change, alongside the automated synthetic-PDF tests, before any change is considered complete.

Chapter 6: Social Impact & Relevance

6.1 Financial Inclusion for Rural Landowners

A large share of Gujarat's agricultural landowners interact with the formal banking system primarily through loans secured against land — loans that depend on a Dastavej or 7/12 extract being read and processed correctly and quickly. Manual processing delays disproportionately affect rural applicants who already have less margin to absorb delay, and every delay is currently a function of document volume, not document importance.

6.2 Reducing Dependence on Intermediaries

Complex, semi-legible legal documents create a natural market for intermediaries who charge a fee to “read and interpret” a Dastavej on someone's behalf. A tool that gives a valuer or bank officer direct, structured access to the same document reduces the informational advantage that sustains that intermediary layer — without removing the human reviewer, who remains firmly in the loop.

6.3 Removing the Language Barrier Between a Village Document and a City Bank

Most mainstream document-AI tooling is built and evaluated against English business documents, on the assumption that whoever needs the extracted data can also read the source document. That assumption breaks down for exactly the documents this project targets: a Dastavej registered in a Gujarati-speaking village frequently needs to be understood by a bank's centralized loan processing team, an out-of-state legal reviewer, or a head office — none of whom can be assumed to read Gujarati script fluently.

By reading the original Gujarati document natively (Chapter 3.2) and standardizing every extracted value into professional English before it reaches a report (Chapter 4.3), the system removes a language barrier that would otherwise require a bilingual intermediary for every single document — reinforcing the same reduction in intermediary dependence discussed in 6.2, on a language basis rather than only a legal-literacy basis.

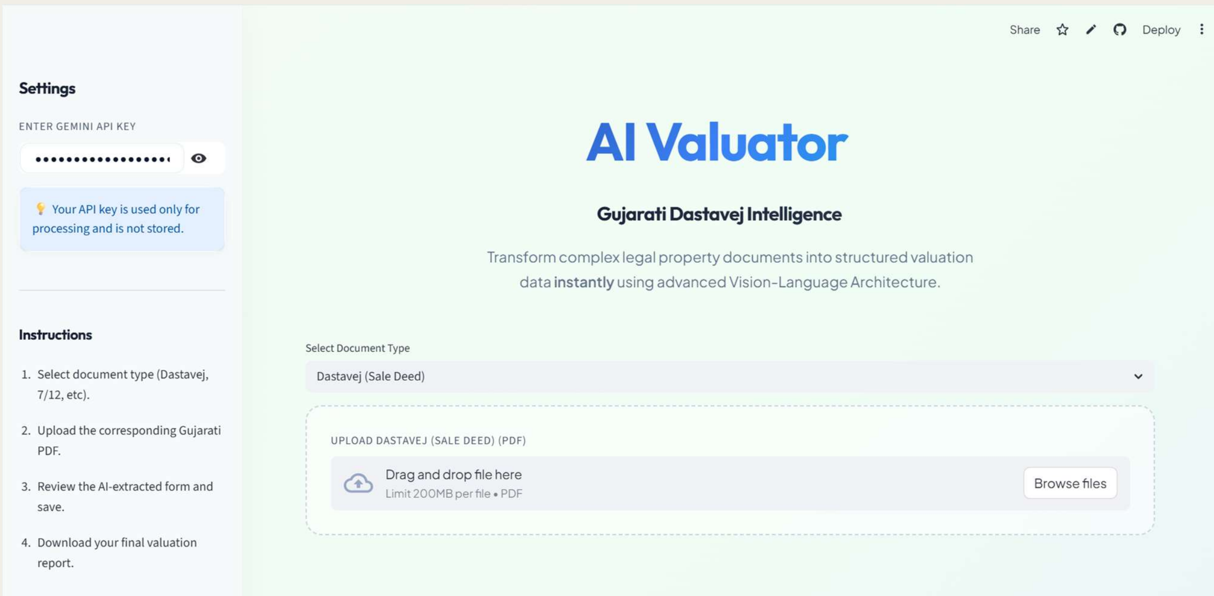
6.4 Fraud and Discrepancy Detection

The AnyROR verification link (Chapter 4.5) exists specifically to make it easy for a reviewer to catch discrepancies — a mismatched owner name, a survey number under an unresolved mutation entry — between what a Dastavej claims and what the government's own live record currently shows. This is precisely the kind of check that protects a bank or buyer from a stale or disputed title, and it is strengthened, not weakened, by keeping a human in that verification loop.

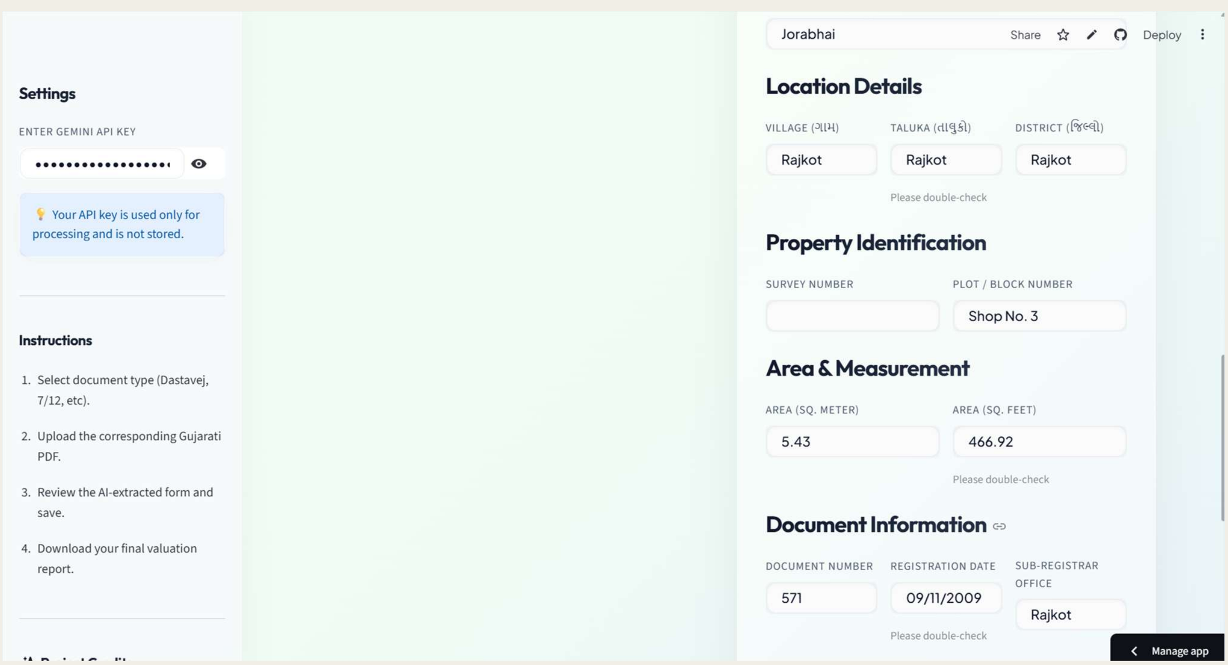
Chapter 7: Results

This section presents the working application. Screenshot placeholders below are marked for insertion — replace each with an actual capture from the live application before final submission; a working demo is available at the live link in Appendix A.

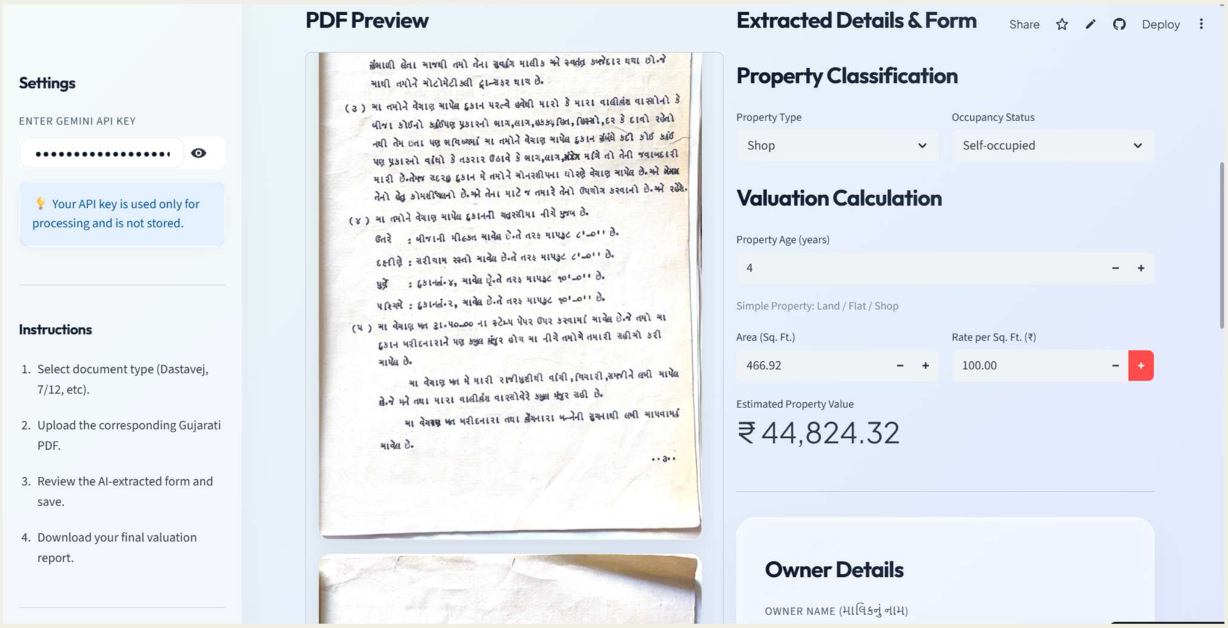
7.1 Document Upload & Type Selection



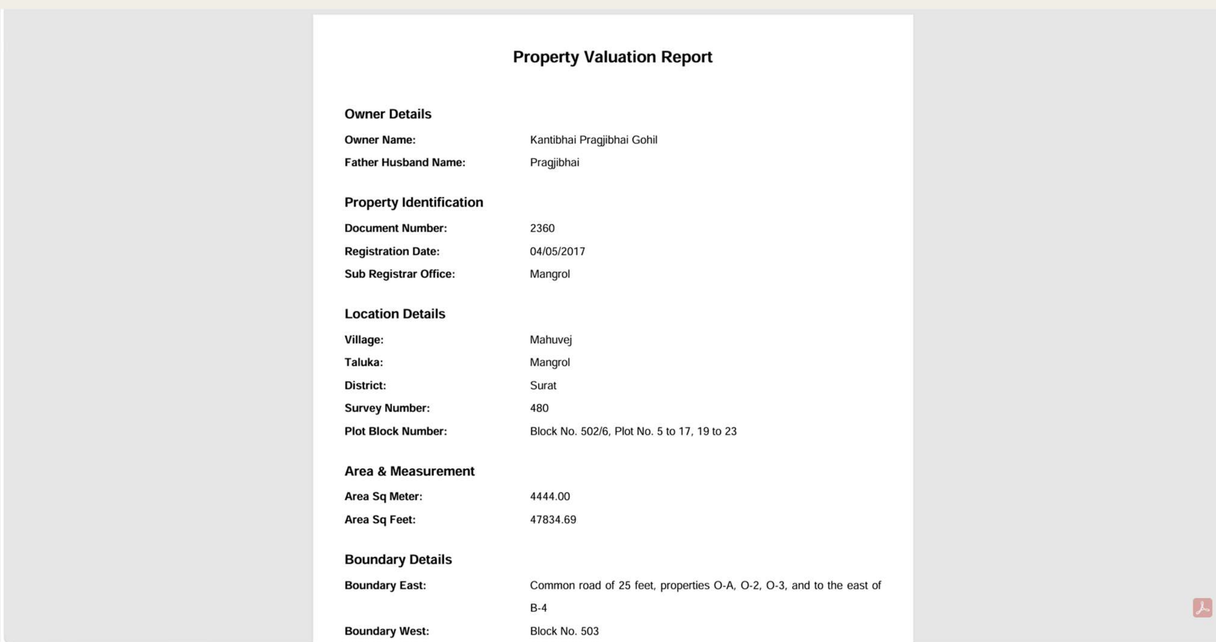
7.2 Confidence-Flagged Review Form



7.3 Live Valuation Calculator



7.4 Professional English PDF Report



Property Valuation Report

Owner Details

Owner Name:	Kantibhai Pragjibhai Gohil
Father Husband Name:	Pragjibhai

Property Identification

Document Number:	2360
Registration Date:	04/05/2017
Sub Registrar Office:	Mangrol

Location Details

Village:	Mahuvej
Taluka:	Mangrol
District:	Surat
Survey Number:	480
Plot Block Number:	Block No. 502/6, Plot No. 5 to 17, 19 to 23

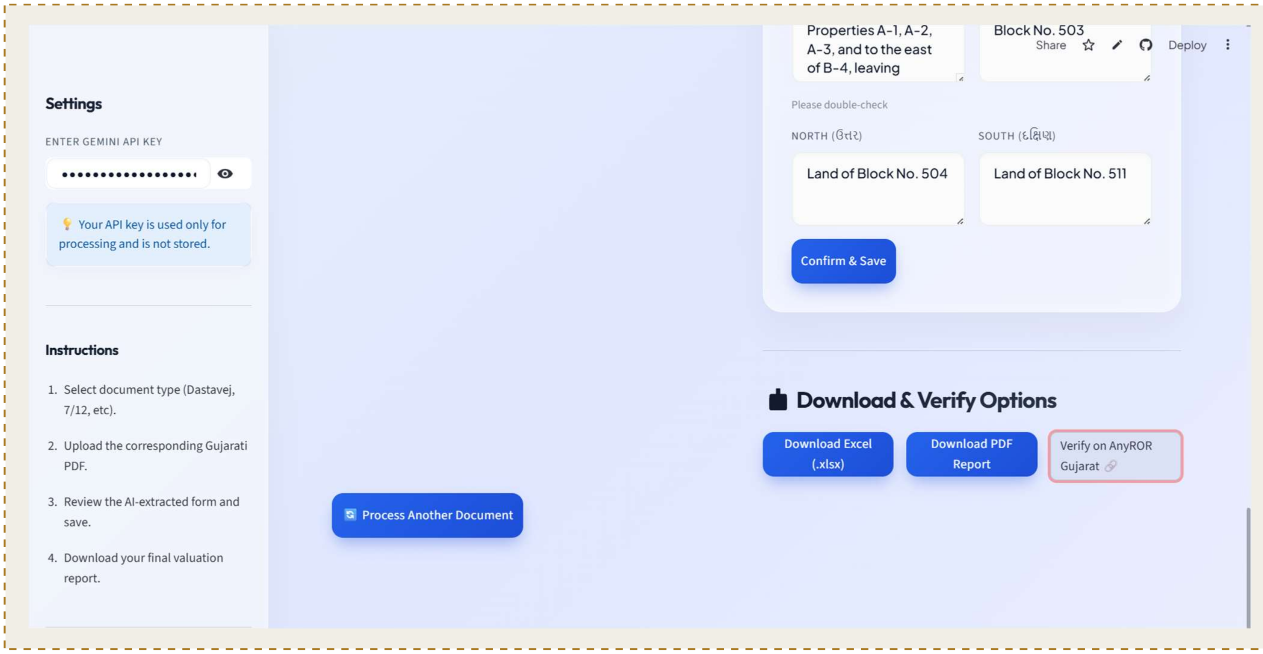
Area & Measurement

Area Sq Meter:	4444.00
Area Sq Feet:	47834.69

Boundary Details

Boundary East:	Common road of 25 feet, properties O-A, O-2, O-3, and to the east of B-4
Boundary West:	Block No. 503

7.5 AnyROR Verification Link



Chapter 8: Conclusion & Future Scope

8.1 Conclusion

The Gujarati Dastavej AI Valuator demonstrates that a vision-language model can reliably read the specific class of document — semi-legible, Gujarati-script, inconsistently formatted — that classical OCR handles poorly, and that this can be delivered as a working, tested tool rather than a proof-of-concept demo. The system extracts data across four real Gujarat land-record formats, flags its own uncertainty, standardizes its output into the professional English institutions already expect, correctly enforces the statutory boundary on which document types can even receive a computed valuation, and gives reviewers a legitimate path to verify extracted data against the government's own record — all while keeping a human reviewer in control of every value before it is used.

Just as importantly, the system was hardened through real, adversarial testing that caught and closed genuine bugs — including a critical rendering failure and a silent calculation error — rather than being presented as complete without that scrutiny.

8.2 Future Scope

The following are planned but not yet implemented, and are listed here rather than presented as delivered features:

- Local persistence (a lightweight database of past valuations, searchable by owner or survey number, with duplicate-upload detection).
- Batch upload and processing for multiple documents of the same type in one pass.
- Server-side API key management with basic passcode-gated access, removing the need for each user to supply their own Gemini API key.
- Camera-capture input for field officers photographing physical documents on-site.
- Integration of Jantri (government guideline rate) values by village/taluka, to move from a user-supplied rate to a government-benchmarked estimate.

References

[1] Google AI for Developers — Gemini API documentation and pricing. https://ai.google.dev/gemini-api/docs

[2] PyMuPDF (fitz) documentation — PDF processing library. https://pymupdf.readthedocs.io/

[3] fpdf2 documentation — PDF generation and Unicode font embedding. https://py-pdf.github.io/fpdf2/

[4] Streamlit documentation. https://docs.streamlit.io/

[5] Government of Gujarat, Revenue Department — AnyROR land record portal. https://anyror.gujarat.gov.in/

[6] Government of Gujarat — Garvi (registration, valuation & indexing) portal. https://garvi.gujarat.gov.in/

[7] Landeed — Gujarat land record retrieval service, referenced for cost comparison. https://web.landeed.com/state/gujarat

[8] Docsumo, Nanonets, Google Document AI — public pricing pages, referenced for cost comparison in Chapter 2.

Appendix A: Quick-Access QR Codes

The following codes provide direct access to the live application, a recorded video demonstration, and the source code repository. All three are functional and scan directly to the destinations listed below.

Live Application
(functional — scan to open)

Video Demonstration
(functional — scan to watch)

Source Code (GitHub)
(functional — scan to view)

Live application: https://valuation-ibedobncjwqw9jth8jcrds.streamlit.app/

Video-demonstration: https://www.dropbox.com/scl/fi/d31b46hmptt2khqcyl72z/Working-Demo.mp4?rlkey=2g8og668ngqnxzswhcrcfngpw&st=t08vyg3y&dl=0

Source code repository: https://github.com/DevanshGanatra/valuation

Appendix B: Field Validation Letter

The letter below was provided by Harsh Kathrani, Technical Manager at Dot Value Consultancy, a practising civil engineer who reviewed and tested this system against a real Dastavej document as described in Chapter 5.2. It is reproduced here as submitted on the firm's letterhead.

Dot Value Consultancy

.Value
Accuracy Builds Credibility

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Date: 5th July 2026

To Whom It May Concern,

Subject: Professional Feedback — Gujarati Dastavej AI Valuator

I am Harsh Kathrani, Technical Manager at Dot Value Consultancy and a Civil Engineer (B.E., Gujarat Technological University). I currently worked as an Assistant Valuer handling market valuation assignment across the Saurashtra and Kutch region, and continue to work with Gujarati property documents, including Dastavej (Sale Deed), Property Card, and Village Form No. 2, in the course of my practice.

Devansh Ganatra shared an early working version of the Gujarati Dastavej AI Valuator with me for review. I tested it against sample documents from my own reference material. In this initial test, the tool correctly identified the general structure of the documents, but the owner's name, the four-side boundary details, and the area and city fields were not extracted accurately and required correction. I shared this feedback directly, and the subsequent version I was shown had addressed these specific issues.

I also discussed the property valuation methodology with him — specifically, two standard approaches used in practice: a composite rate method, where value is calculated as land area multiplied by rate; and a land-and-building method, where land value (area × rate) and built-up construction value are calculated separately and added together for the total property value. I understand both methods were incorporated into the tool's valuation calculation as I described them.

My review and the comments in this letter are limited to the document types and valuation methods described above, which fall within a Civil Engineer's professional domain. Land records or valuation matters that fall under a different technical or regulatory authority are outside the scope of my professional practice, and this letter should not be read as an opinion on those.

In my professional opinion, this tool addresses a genuine and everyday burden in how Gujarati property documents are currently processed — manual reading and transcription of semi-legible, regional-language documents is slow and error-prone in practice. What I have seen shows real promise as a productivity aid for valuers and lending institutions. I would recommend further testing across a larger and more varied set of real documents before it is relied upon in a live lending or legal workflow, and I would continue to treat its output as a preliminary draft for a certified valuer's review rather than a substitute for one.

Sincerely,

Harsh Kathrani
Technical Manager, Dot Value Consultancy

Figure: Field Validation Letter — Dot Value Consultancy (Harsh Kathrani, Technical Manager)